Grade 11 - Term 1
C1–C4 Practice Exercises and Solutions

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Evaluated criteria

C1 Characterises continuous random variables by distinguishing them from discrete variables and interpreting their probability density function, cumulative distribution function, and support in economic contexts.

C2 Applies uniform and triangular continuous distributions with constant or piecewise-linear probability density functions to calculate interval probabilities using geometric areas.

C3 Applies circular-arc probability density functions to calculate interval probabilities using the geometry of circles and circular segments.

C4 Applies the exponential distribution to constant-rate waiting-time situations and assesses its suitability.

1 Identifying Probability Distributions from Contexts [C1-5]

A variety of Colombian organizations use probability models to represent uncertain quantities and events. In each situation below, a random variable is defined from the information provided. The appropriate probability distribution must be identified from the characteristics of the situation.

Questions/Tasks

C1 For each context, characterize the random variable and identify the appropriate probability distribution. **[5]**

Context 1. A university sends an invitation to exactly 60 randomly selected alumni to participate in a fundraising campaign. Each alumnus either makes a donation or does not. The university records the number of alumni who donate to estimate the expected response to the campaign.

Context 2. A courier company models the travel time for a delivery route. Travel times range from 25 to 55 minutes, with times around 40 minutes occurring most frequently. Values become progressively less common as they move away from 40 minutes toward either endpoint. The company uses this model to estimate delivery durations.

Context 3. A telecommunications company counts the number of incoming technical-support requests received by a service center during a fixed 30-minute period. Historical records indicate that requests occur independently and without a predictable pattern. The company uses this model to plan the number of available support agents.

Context 4. A water-utility company studies the daily water consumption of a small residential building. Consumption ranges from 100 to 500 litres, and the probability density increases steadily as consumption approaches 500 litres. The company uses this model to plan daily supply capacity.

Context 5. A manufacturer examines batches of circuit boards until the first board that fails a functional test is found. Boards are tested independently, and testing stops immediately after the first failure. The manufacturer records the number of boards tested, including the failed board, to estimate inspection requirements.

Context 6. A logistics company studies the delay, in minutes, of trucks arriving at a distribution center. Delays can take any value between 0 and 30 minutes, with all values in this interval considered equally likely. The company uses this model to estimate scheduling margins.

Context 7. A hospital records the number of patients arriving at an emergency department during each fixed four-hour shift. Comparable shifts show that arrivals occur independently and without a predictable timetable. The hospital uses this model to plan staffing levels.

Context 8. A cinema chain sends a discount code to exactly 150 randomly selected customers. Each customer either uses the code or does not. The chain records the number of customers who use the discount to estimate the additional attendance generated by the campaign.

Context 9. A dairy producer measures the fat content of milk samples from its regular production process. Measurements cluster around 3.5%. Measurements farther from 3.5% occur progressively less often. The producer uses this model to establish quality-control limits.

Context 10. A city's electric utility studies the daily electricity consumption of a small commercial building. Consumption ranges from 200 to 800 kWh, and the probability density decreases steadily as consumption moves from 200 kWh toward 800 kWh. The utility uses this model to estimate the likelihood of different daily loads.

C1 - Distribution identification: Solutions [5]

Model qualification. The labels below are the intended candidate models, rather than uniquely determined distributions. Contexts 1 and 8 need independent trials with a common success probability; Context 5 needs a constant failure probability; Contexts 3 and 7 need a constant arrival rate for a Poisson model. Peaked or monotone descriptions in Contexts 2, 4, 9 and 10 do not alone establish triangular, linear or normal densities. In particular, Context 9 does not explicitly state symmetry or a normal shape.

1. **Context 1 — Discrete, Binomial.** X_1 counts donations among 60 selected alumni, with each alumnus having two possible outcomes.
2. **Context 2 — Continuous, Triangular.** X_2 has a minimum of 25 minutes, maximum of 55 minutes, and mode of 40 minutes.
3. **Context 3 — Discrete, Poisson.** X_3 counts independent technical-support requests occurring during a fixed time interval.
4. **Context 4 — Continuous, Linear.** X_4 has a density that increases steadily toward 500 litres.
5. **Context 5 — Discrete, Geometric.** X_5 counts the number of boards tested until the first failure.
6. **Context 6 — Continuous, Uniform.** X_6 can take any value in $[0, 30]$, with all values equally likely.
7. **Context 7 — Discrete, Poisson.** X_7 counts independent patient arrivals during a fixed period.
8. **Context 8 — Discrete, Binomial.** X_8 counts discount-code users among 150 customers, with each customer having two possible outcomes.
9. **Context 9 — Continuous, Normal.** X_9 measures fat content in percent, clustered around 3.5%; a normal model is a candidate if an approximately symmetric bell shape is confirmed.
10. **Context 10 — Continuous, Linear.** X_{10} has a density that decreases steadily from 200 kWh toward 800 kWh.

2 Charging Time for an Electric Vehicle [C2-3]

A transportation company studies the additional charging time required for electric vehicles at one of its charging stations. For a particular type of vehicle, the additional charging time can range from 0 to 4 minutes. A charging time of any length within this interval is possible, and shorter additional times become steadily more common. Historical data indicate that an additional charging time of 0 minutes is twice as common as an additional charging time of 4 minutes.

Questions/Tasks

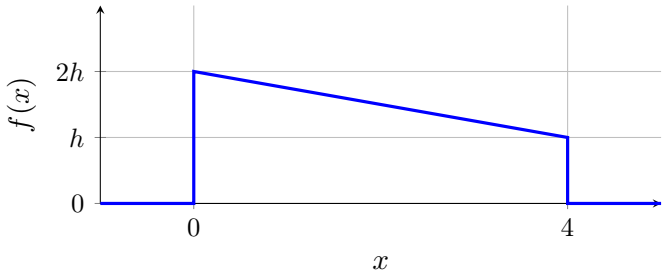
C2 Determine the probability that the additional charging time is between 1 and 3 minutes. **[3]**

C2 - Linear probability distribution [3]

2.1 Support and Shape

The random variable X has support $[0, 4]$. Under the intended linear model, the density decreases linearly, and its height at 0 is twice its height at 4. Let the endpoint heights be $2h$ and h. Here "twice as common" must refer to density: exact values of a continuous random variable each have probability zero. Also, "steadily" does not mathematically specify linearity; without that intended model, the probability is not uniquely determined.

Shape from the context



2.2 Total Area = 1

The total area under a probability density function is 1. For a trapezoid with parallel sides of heights h_1 and h_2 and base b,

$$A = \frac{h_1 + h_2}{2}(b)$$

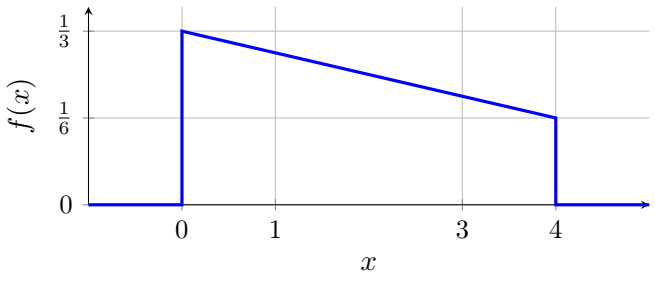
Therefore, in this case,

$$1 = \frac{2h + h}{2}(4) \qquad \rightarrow \qquad 1 = 3h(2) \qquad \rightarrow \qquad h = \frac{1}{6}.$$

Therefore,

$$f(0) = 2h = \frac{1}{3}, \qquad f(4) = h = \frac{1}{6}.$$

Completed density graph



2.3 Complete pdf

The density is the straight line through

$$(x_1, y_1) \qquad (x_2, y_2)$$
$$\left(0, \frac{1}{3}\right) \qquad \left(4, \frac{1}{6}\right)$$

For a general straight line,

$$f(x) = mx + b \qquad \text{where} \qquad m = \frac{y_2 - y_1}{x_2 - x_1} \qquad \text{and} \qquad b = y_1 - mx_1$$

Substituting the given points,

$$m = \frac{\frac{1}{6} - \frac{1}{3}}{4 - 0} = -\frac{1}{24} \qquad b = \frac{1}{3} - \frac{-1}{24}(0) = \frac{1}{3}$$

$$f(x) = -\frac{x}{24} + \frac{1}{3} = -\frac{x}{24} + \frac{8}{24} = \frac{8 - x}{24}$$

Therefore,

$$f(x) = \begin{cases} \dfrac{8 - x}{24}, & 0 \le x \le 4 \\ 0, & \text{otherwise} \end{cases}$$

The density is nonnegative on $[0, 4]$, zero outside, and $\int_0^4 (8 - x)/24\, dx = 1$; its units are inverse minutes.

2.4 Probability Between 1 and 3 Minutes

The required probability is the area under the density from 1 to 3:

$$P(1 \le X \le 3) = \frac{f(1) + f(3)}{2}(3 - 1)$$

Since

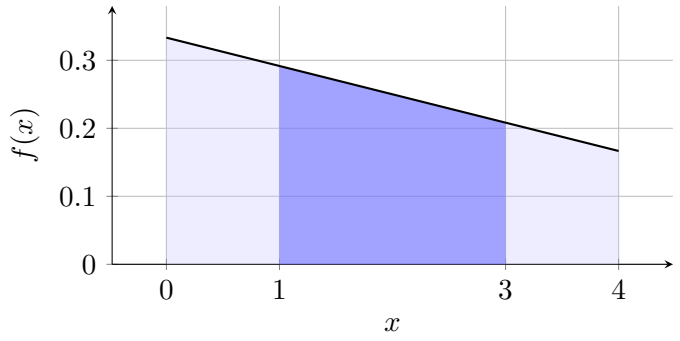
$$f(1) = \frac{8 - 1}{24} = \frac{7}{24}, \qquad f(3) = \frac{8 - 3}{24} = \frac{5}{24}.$$

$$P(1 \le X \le 3) = \frac{1}{2}\left(\frac{7}{24} + \frac{5}{24}\right)(2) = \frac{1}{2}.$$

Therefore,

$$P(1 \le X \le 3) = \frac{1}{2} = 50\% \,.$$

Band graph: $P(1 \le X \le 3)$



3 Daily Sales Revenue of a Local Business [C2-4]

A local business records its daily sales revenue over several months. Daily revenue ranges from \$1 million to \$7 million. Revenue becomes steadily more common from \$1 million up to a most common level of \$6 million, then steadily less common through \$7 million; there are no observations exactly at either limit.

Questions/Tasks

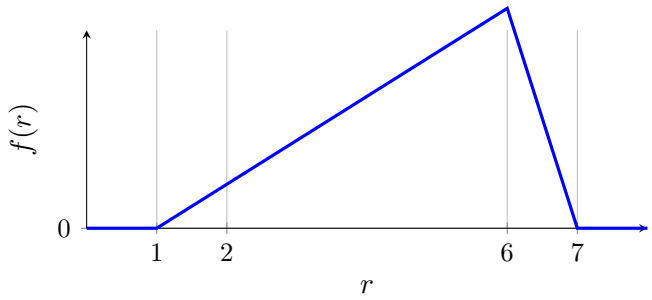
C2 Determine the probability that the daily sales revenue is between \$2 million and \$6 million. **[4]**

C2 - Triangular probability distribution [4]

3.1 Support and Shape

Let R represent the daily sales revenue, measured in millions of dollars. The random variable R has support $[1, 7]$. Under the intended triangular model, its density rises linearly to a peak at $r = 6$, then falls linearly to zero at $r = 7$. Let the peak height be H. The words "steadily" and "no observations exactly at either limit" do not by themselves establish straight segments or zero endpoint density. Thus the numerical answer uses the intended triangular model; the literal statement alone does not uniquely determine it.

Shape from the context



3.2 Total Area = 1

The total area under a probability density function is 1. For a triangle with base b and height H,

$$A = \frac{1}{2}bH.$$

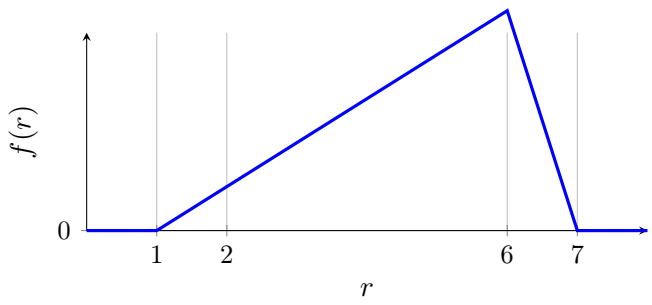
Therefore, in this case,

$$1 = \frac{1}{2}(7-1)H \quad \rightarrow \quad 1 = \frac{1}{2}(6)H \quad \rightarrow \quad 1 = 3H \quad \rightarrow \quad H = \frac{1}{3}.$$

Therefore,

$$f(6) = \frac{1}{3}.$$

Completed density graph



3.3 Complete pdf

The density consists of two straight-line segments. For a general straight line,

$$f(r) = mr + b, \qquad \text{where} \qquad m = \frac{y_2 - y_1}{x_2 - x_1} \qquad \text{and} \qquad b = y_1 - mx_1.$$

For the rising segment, the line passes through

$$\begin{array}{cc} (x_1, y_1) & (x_2, y_2) \\ (1, 0) & \left(6, \frac{1}{3}\right) \end{array}$$

Thus

$$m_1 = \frac{\frac{1}{3} - 0}{6 - 1} = \frac{1}{15} \qquad b_1 = 0 - \frac{1}{15}(1) = -\frac{1}{15}.$$

Therefore,

$$f(r) = \frac{r}{15} - \frac{1}{15} = \frac{r - 1}{15}.$$

For the falling segment, the line passes through

$$\left(6, \frac{1}{3}\right) \qquad \text{and} \qquad (7, 0).$$

However, these calculations are not needed for the requested probability, because the interval $2 \le R \le 6$ ends at the mode. Therefore, the equation of the falling segment is stated directly without showing its derivation.

The complete probability density function is

$$f(r) = \begin{cases} \dfrac{r - 1}{15}, & 1 \le r \le 6, \\ \dfrac{7 - r}{3}, & 6 < r \le 7, \\ 0, & \text{otherwise.} \end{cases}$$

Both branches are nonnegative on their support, and their areas are $5/6 + 1/6 = 1$. Density units are inverse million dollars.

3.4 Probability Between $2 and $6 Million

The complete triangular density has a base extending from $r = 1$ to $r = 7$. The required interval extends from $r = 2$ to the mode at $r = 6$. Therefore, the area outside the required interval consists of two tails: the left tail from $r = 1$ to $r = 2$, and the right tail from $r = 6$ to $r = 7$:

$$P(2 \le R \le 6) = 1 - P(R < 2) - P(R > 6).$$

For the left tail, the density is

$$f(r) = \frac{r-1}{15}.$$

Thus,

$$f(2) = \frac{2-1}{15} = \frac{1}{15}.$$

The left-tail area is therefore

$$P(R < 2) = \frac{1}{2}(2-1)\left(\frac{1}{15}\right) = \frac{1}{30}.$$

At the mode the height is the height of the triangle as seen in part 1.

$$f(6) = \frac{1}{3}$$

The right-tail area is therefore

$$P(R > 6) = \frac{1}{2}(7-6)\left(\frac{1}{3}\right) = \frac{1}{6}.$$

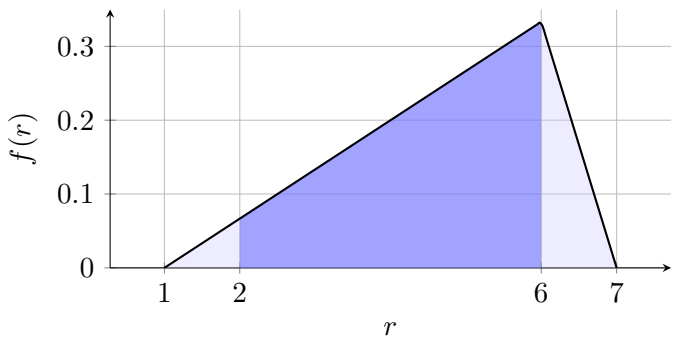
Consequently,

$$\begin{aligned} P(2 \le R \le 6) &= 1 - \frac{1}{30} - \frac{1}{6} \\ &= 1 - \frac{1}{30} - \frac{5}{30} \\ &= 1 - \frac{6}{30} \\ &= \frac{24}{30} \\ &= \frac{4}{5}. \end{aligned}$$

Therefore,

$$P(2 \le R \le 6) = \frac{4}{5} = 80\% \ .$$

Band graph: $P(2 \le R \le 6)$



4 Timing of an Overnight Price Adjustment [C3-2]

A retail company monitors the time of day at which an unexpected price adjustment occurs in one of its online markets. The adjustment is equally likely to occur at any time during a 24-hour period, so no part of the day is more likely than another.

Questions/Tasks

C3 Determine the probability that the price adjustment occurs between 8 p.m. and 6 a.m. **[2]**

C3 - Uniform probability distribution [2]

4.1 Uniform Distribution

Let T represent the time of the price adjustment, measured in hours after midnight. Since every time during the 24-hour period is equally likely,

$$T \sim U(0, 24) \qquad \longrightarrow \qquad f(t) = \frac{1}{24 - 0} = \frac{1}{24}.$$

Here $f(t) = 0$ outside $[0, 24)$, and $\int_0^{24} (1/24)\, dt = 1$; density units are inverse hours.

4.2 Probability Between 8 p.m. and 6 a.m.

The interval crosses midnight, so it consists of

$$20 \le T < 24 \qquad \text{and} \qquad 0 \le T \le 6.$$

The probability is found by calculating the area of each rectangle individually and then adding them:

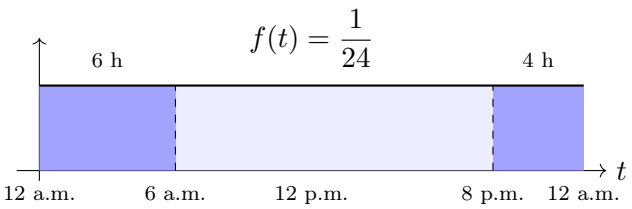
$$A_1 = (6 - 0)\left(\frac{1}{24}\right) = \frac{6}{24},$$

$$A_2 = (24 - 20)\left(\frac{1}{24}\right) = \frac{4}{24}.$$

Therefore,

$$P(\text{between 8 p.m. and 6 a.m.}) = A_1 + A_2 = \frac{6}{24} + \frac{4}{24} = \frac{10}{24} = \boxed{\frac{5}{12} \approx 41.67\%}.$$

Probability between 8 p.m. and 6 a.m.



5 Direction of an Unexpected Sensor Reading [C3-3]

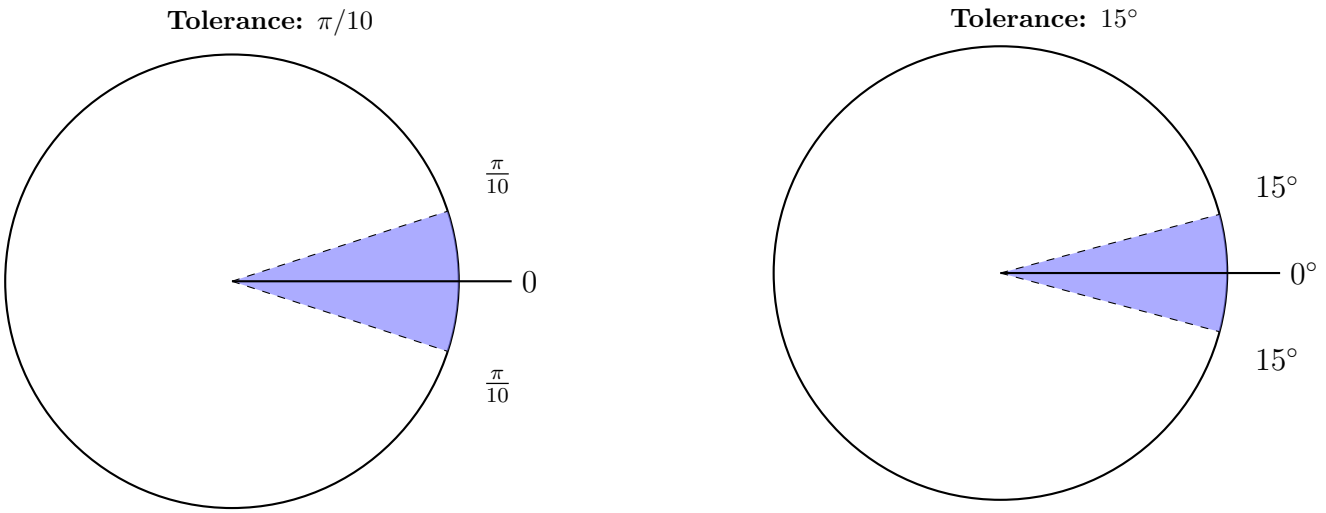
A directional sensor records the direction in which it is pointing. The sensor can point in any direction around a complete circle, and every direction is equally likely. Let A represent the direction of the sensor, measured in radians from a fixed reference marker at 0 radians.

Questions/Tasks

C3 Determine which is more probable and by how much: the sensor pointing within $\pi/10$ radians of the reference marker, or the sensor pointing within $15°$ of the reference marker. **[3]**

C3 - Uniform probability distribution [3]

Directions within a range around the reference marker



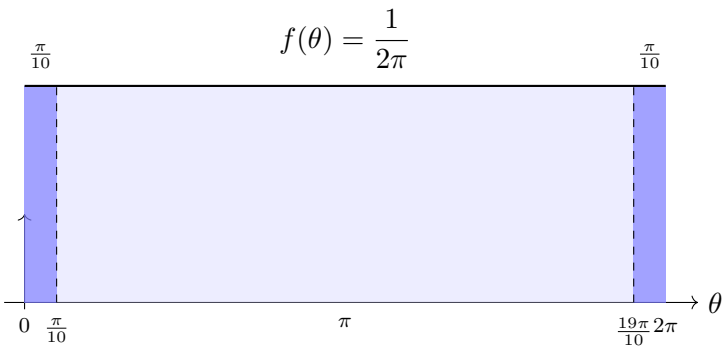
For the angle in radians, $f_A(a) = 1/(2\pi)$ on $[0, 2\pi)$ and zero elsewhere; its integral is one. The degree-labelled plot below uses radian coordinates, so its density height is also $1/(2\pi)$.

5.1 Probability Within $\pi/10$ of the Reference Marker

The sensor can point uniformly around a complete circle of 2π radians. Being within $\pi/10$ radians of the reference marker includes the same angular distance on both sides of the marker, so the total angular width is

$$2\left(\frac{\pi}{10}\right) = \frac{\pi}{5}. \qquad \longrightarrow \qquad P\left(\text{within } \frac{\pi}{10} \text{ of the reference marker}\right) = \frac{\pi/5}{2\pi} = \boxed{\frac{1}{10} = 10\%}.$$

Probability Within $\pi/10$ of the Reference Marker

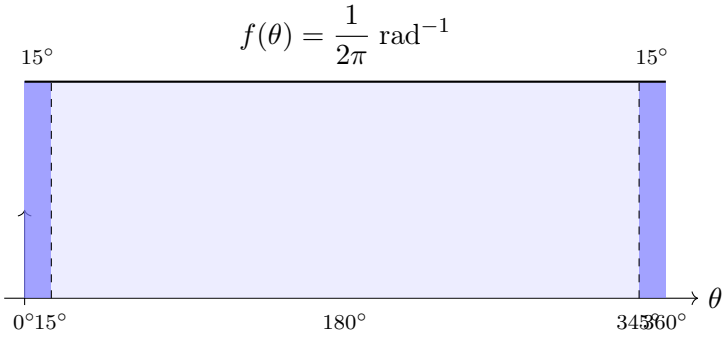


5.2 Probability Within 15° of the Reference Marker

A complete circle measures 360°. Being within 15° of the reference marker gives a total angular width of

$$2(15^\circ) = 30^\circ. \qquad \longrightarrow \qquad P(\text{within } 15^\circ \text{ of the reference marker}) = \frac{30^\circ}{360^\circ} = \boxed{\frac{1}{12} \approx 8.33\%}.$$

Probability Within 15° of the Reference Marker



5.3 Comparison

Since

$$\frac{1}{10} > \frac{1}{12}, \qquad \longrightarrow \qquad \frac{1}{10} - \frac{1}{12} = \frac{1}{60} \approx 0.0167.$$

Thus,

Within $\frac{\pi}{10}$ radians is more probable than within 15° by $\frac{1}{60}$ (1.67 percentage points).



Exponential model check (Problems 6–12). For $\lambda > 0$, $f_T(t) = \lambda e^{-\lambda t}$ on $[0, \infty)$ and zero otherwise is nonnegative, with $\int_0^\infty \lambda e^{-\lambda t}\, dt = 1$. Rates below are per minute and waiting times are in minutes. Problems 6–8 supply average rates but do not explicitly give a constant hazard or independent arrivals; their answers use the intended exponential model. Expected counts in Problem 9 use the standard homogeneous Poisson interpretation of independent exponential interarrival times.

6 Waiting for a Food-Delivery Driver [C4-0.5]

A food-delivery platform receives driver arrivals at an average rate of 5 drivers every 8 minutes.

Questions/Tasks

C0.5 Determine the probability that the next driver arrives more than 2 minutes after the start of the observation. **[0.5]**

C4 - Exponential probability distribution [0.5]

The average rate is 5 drivers every 8 minutes, so

$$\lambda = \frac{5}{8} = 0.625.$$

General case: $P(T > t) = e^{-\lambda t}$.

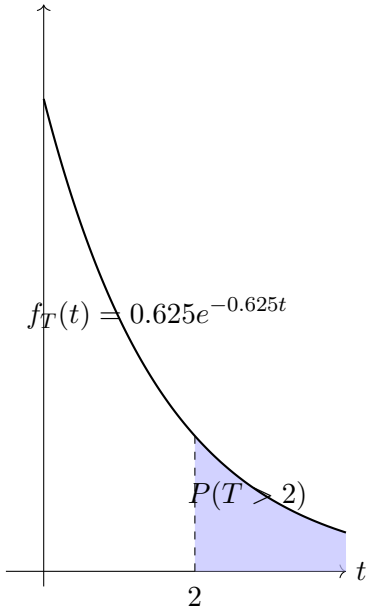
Thus,

$$P(T > 2) = e^{-(0.625)(2)} = e^{-1.25} \approx 0.2865.$$

Therefore,

$$P(T > 2) \approx 28.65\%.$$

Probability region



7 Waiting for a Public-Transport Delay Alert [C4-0.5]

A public-transport monitoring system detects significant service delays at an average rate of 4 alerts every 15 minutes.

Questions/Tasks

C0.5 Determine the probability that the next significant delay is detected within 3 minutes after the start of the observation. **[0.5]**

C4 - Exponential probability distribution [0.5]

The average rate is 4 significant delays every 15 minutes, so

$$\lambda = \frac{4}{15} \approx 0.2667.$$

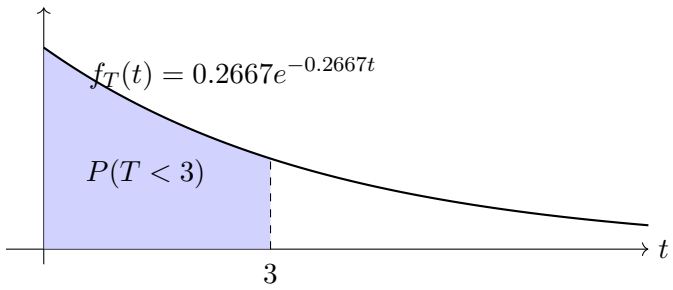
General case: $P(T < t) = 1 - e^{-\lambda t}$.

$$P(T < 3) = 1 - e^{-(4/15)(3)} = 1 - e^{-4/5} \approx 0.5507.$$

Therefore,

$$P(T < 3) \approx 55.07\%.$$

Probability region



8 Waiting for a Supply Chain Disruption Alert [C4-1]

A supply chain monitoring system detects significant disruptions at an average rate of 5 alerts every 20 minutes.

Questions/Tasks

C1 Determine the probability that the next significant disruption is detected between 4 and 8 minutes after the observation time. **[1]**

C4 - Exponential probability distribution [1]

The average rate is 5 significant disruptions every 20 minutes, so

$$\text{In general} \qquad P(a < T < b) = e^{-\lambda a} - e^{-\lambda b} \qquad \text{where} \qquad \lambda = \frac{5}{20} = \frac{1}{4}.$$

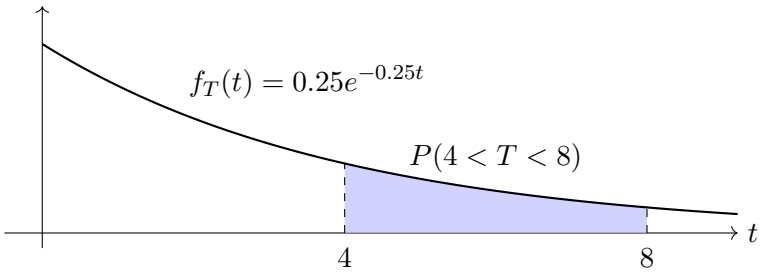
Thus

$$P(4 < T < 8) = e^{-\left(\frac{1}{4}\right)(4)} - e^{-\left(\frac{1}{4}\right)(8)} = e^{-1} - e^{-2} \approx 0.2325.$$

Therefore,

$$P(4 < T < 8) \approx 23.25\%$$

Probability region



9 Monitoring Online Store Orders [C4-1]

An online store receives customer orders throughout the day. The time between consecutive orders is modeled by an exponential distribution. Based on previous observations, there is a 60% probability that the next order is received within 5 minutes.

Questions/Tasks

C1 On average, how many orders are expected to be received in a 12-minute interval? **[1]**

C4.9 - Number of orders

There is a 60% probability that the next order is received within 5 minutes. Using

$$P(T < t) = 1 - e^{-\lambda t},$$

we obtain

$$0.60 = 1 - e^{-5\lambda} \quad \longrightarrow \quad \lambda = -\frac{\ln(0.40)}{5} \approx 0.1833 \ \frac{\text{orders}}{\text{minute}}$$

Therefore, over a 12-minute interval, the order rate gives

$$E[N(12)] = 12\left(-\frac{\ln(0.40)}{5}\right) \approx \boxed{2.1991 \text{ orders}}$$

10 Waiting for an Economic Indicator Update [C4-1]

An economic analysis system monitors market indicators throughout the day. The time between consecutive significant indicator updates is modeled by an exponential distribution. Based on previous observations, there is a 25% probability that the next significant update takes more than 10 minutes to occur.

Questions/Tasks

C1 Find the average number of significant updates expected per minute. **[1]**

C4.9 - Average number of updates per minute

There is a 25% probability that a significant economic indicator update takes more than 10 minutes to occur. Using

$$P(T > t) = e^{-\lambda t},$$

we obtain

$$0.25 = e^{-10\lambda} \quad \longrightarrow \quad \lambda = -\frac{\ln(0.25)}{10} \approx \boxed{0.1386 \ \frac{\text{updates}}{\text{minute}}}$$

11 Comparing Economic Forecasting Rates [C4-2]

wo economic forecasting systems monitor significant changes in market indicators throughout the day. The detected changes by each system are modeled by independent Poisson processes.

For System A, there is a 70% probability that the next significant change is detected within 10 minutes. For System B, there is a 50% probability that the next significant change is detected within 6 minutes.

Questions/Tasks

C2 Determine after how many minutes the expected numbers of significant changes detected by the two systems differ by 1 change. **[2]**

C4.10 - Comparing detection rates

For an exponential waiting time T, if

$$P(T < t) = p \qquad \longrightarrow \qquad p = 1 - e^{-\lambda t} \qquad \longrightarrow \qquad \lambda = -\frac{\ln(1-p)}{t}$$

For System A, $P(T < 10) = 0.70$, so

$$\lambda_A = -\frac{\ln(1-0.70)}{10} \approx 0.1204 \ \frac{\text{changes}}{\text{minute}}$$

For System B, $P(T < 6) = 0.50$, so

$$\lambda_B = -\frac{\ln(1-0.50)}{6} \approx 0.1155 \ \frac{\text{changes}}{\text{minute}}$$

Therefore, System A has the higher detection rate. The difference between the rates is

$$\lambda_A - \lambda_B = -\frac{\ln(0.30)}{10} + \frac{\ln(0.50)}{6} \approx 0.0048727503 \ \frac{\text{changes}}{\text{minute}}$$

After t minutes, this rate difference accumulates to $(\lambda_A - \lambda_B)t$ changes. We want this difference to reach 1 change:

$$(\lambda_A - \lambda_B)t = 1. \qquad \longrightarrow \qquad t = \frac{1}{\lambda_A - \lambda_B} = \frac{1}{-\ln(0.30)/10 + \ln(0.50)/6} \approx \boxed{205.2229 \text{ minutes}}$$

12 Finding a 50% Probability Interval for an Exponential Distribution [CX-2]

This problem can be used to add 2 points for any criteria

The waiting time between successive economic events is modeled by an exponential distribution with rate $\lambda > 0$ events per minute. Suppose we begin observing the system after it has already been running for 1 minute. We do not know whether the next event has already occurred during that first minute.

Questions/Tasks

C2 From the moment we begin observing, how much longer should we wait to have a (50%) chance of seeing the alarm go off? Express this waiting time in terms of λ. Is there any restriction on the value of λ? If so, why? **[2]**

C4.10 - Finding a 50% probability interval

Ambiguity in the exam. The statement does not specify whether the alarm concerns only the first event measured from time zero, or any subsequent event in a recurring process. These give different answers.

Single-event interpretation (unconditional). If an alarm that already occurred is missed permanently, the requested probability is $P(1 < T \leq 1 + L) = 1/2$. The calculation below applies to this interpretation.

For an exponential random variable with rate λ,

$$f(t) = \lambda e^{-\lambda t}, \qquad t \geq 0,$$

and its cumulative distribution function is

$$F(t) = P(T \leq t) = 1 - e^{-\lambda t}.$$

Let the interval begin at $t = 1$ and end at $t = b$. Using the cumulative distribution function, we want

$$F(b) - F(1) = P(1 \leq T \leq b) = 0.50.$$

Substituting $F(t) = 1 - e^{-\lambda t}$,

$$\left(1 - e^{-\lambda b}\right) - \left(1 - e^{-\lambda}\right) = 0.50.$$

Simplifying,

$$e^{-\lambda} - e^{-\lambda b} = 0.50.$$

Therefore,

$$e^{-\lambda b} = e^{-\lambda} - 0.50.$$

Taking the natural logarithm,

$$-\lambda b = \ln\left(e^{-\lambda} - 0.50\right). \qquad \longrightarrow \qquad b = -\frac{1}{\lambda}\ln\left(e^{-\lambda} - 0.50\right)$$

Since the interval starts at 1, its length is

$$L = b - 1.$$

Therefore,

$$\boxed{L = -\frac{1}{\lambda}\ln\left(e^{-\lambda} - 0.50\right) - 1}.$$

For this interval to contain 50% of the probability, we require

$$e^{-\lambda} > 0.50,$$

which gives

$$\boxed{\lambda < \ln 2}$$

For a finite waiting time, the full condition is $0 < \lambda < \ln 2$ (with time measured in minutes). At $\lambda = \ln 2$, the probability approaches $1/2$ only as $L \to \infty$; for larger rates it never reaches $1/2$.

Recurring-event interpretation. If successive waiting times are independent exponentials (a homogeneous Poisson process), the wait W from observation to the next event satisfies

$$P(W \leq L) = 1 - e^{-\lambda L} = \tfrac{1}{2}, \qquad \boxed{L = \frac{\ln 2}{\lambda}}.$$

This holds for every $\lambda > 0$, even if events occurred in the first minute. The same formula holds for a single event conditional on $T > 1$, but that conditioning is not given in the exam. Clarification is required to select a unique answer.

13 Summarized Solutions [C1-C4]

13.1 Identifying Probability Distributions from Contexts C1–5

These are candidate models subject to the qualifications in the detailed solution.

Context	Class.	Distribution	Context	Class.	Distribution
1	D	Binomial	6	C	Uniform
2	C	Triangular	7	D	Poisson
3	D	Poisson	8	D	Binomial
4	C	Linear	9	C	Normal
5	D	Geometric	10	C	Linear

13.2 Charging Time for an Electric Vehicle C2–3

Under the intended linear model:

$$f(x) = \begin{cases} \dfrac{8-x}{24} & 0 \le x \le 4 \\ 0 & \text{otherwise} \end{cases}$$

$$P(1 \le X \le 3) = \frac{1}{2} = 50\%$$

13.3 Daily Sales Revenue of a Local Business C2–4

Under the intended triangular model:

$$f(r) = \begin{cases} \dfrac{r-1}{15} & 1 \le r \le 6 \\ \dfrac{7-r}{3} & 6 < r \le 7 \\ 0, & \text{otherwise} \end{cases}$$

$$P(2 \le R \le 6) = \frac{4}{5} = 80\%$$

13.4 Timing of an Overnight Price Adjustment C3–2

$$T \sim U(0, 24)$$

$$P(20 \le T < 24 \text{ or } 0 \le T \le 6) = \frac{5}{12} \approx 41.67\%$$

13.5 Direction of an Unexpected Sensor Reading C3–3

$$P\left(\text{within } \frac{\pi}{10} \text{ of the marker}\right) = \frac{1}{10}$$

$$P\left(\text{within } 15^\circ \text{ of the marker}\right) = \frac{1}{12}$$

Therefore, the $\pi/10$-radian tolerance is more probable by

$$\frac{1}{60} \text{ (1.67 percentage points)}$$

13.6 Waiting for a Food-Delivery Driver C4–0.5

$$\lambda = \frac{5}{8},$$

$$P(T > 2) = e^{-1.25} \approx 28.65\% .$$

13.7 Waiting for a Public-Transport Delay Alert C4–0.5

$$\lambda = \frac{4}{15},$$

$$P(T < 3) = 1 - e^{-0.8} \approx 55.07\% .$$

13.8 Waiting for a Supply Chain Disruption Alert C4–1

$$\lambda = \frac{1}{4},$$

$$P(4 < T < 8) = e^{-1} - e^{-2}$$

$$P(4 < T < 8) \approx 23.25\% .$$

13.9 Monitoring Online Store Orders C4–1

$$\lambda = -\frac{\ln(0.40)}{5} \approx 0.1833$$

$$E[N(12)] \approx 2.1991 \text{ orders}$$

13.10 Waiting for an Economic Indicator Update C4–1

$$\lambda = -\frac{\ln(0.25)}{10} \approx 0.1386 \ \frac{\text{updates}}{\text{minute}}$$

13.11 Comparing Economic Forecasting Rates C4–2

$$\lambda_A \approx 0.1204 \qquad \lambda_B \approx 0.1155$$

System A has the higher rate; the expected counts differ by one after

$$t \approx 205.2229 \text{ minutes}$$

13.12 Finding a 50% Probability Interval CX–2

$$b = -\frac{1}{\lambda} \ln\left(e^{-\lambda} - 0.50\right)$$

$$L = b - 1 = -\frac{1}{\lambda} \ln\left(e^{-\lambda} - 0.50\right) - 1$$

For a single event, a finite wait requires

$$0 < \lambda < \ln 2$$

For a recurring Poisson process, instead:

$$L = \ln 2/\lambda, \quad \lambda > 0 .$$

The exam does not select a unique interpretatior